



Ripha International University (Sahiwal Campus) Department of Computing

Course Title and Code	Calculus and Analytical Geometry	Credits	3 (3+0)
Instructor	Shumaila Asghar		
Exam Type	Mid Term		
Program	BSIT		
Session	Fall-2025		
Semester	2 nd		
Time Allowed	10 minutes		
Maximum Marks	6		
Student's Name		Roll No.	Section
CLO	Question No	Marks	Obtained
01	1,3	14	
02	2,4	16	

Instructor's Signatures: _____ Course Moderator/HOD's Signatures: _____

(Objective)

Question # 1 Choose the correct option.

I. The set of all possible input values of a function is called

- a) Domain b) Range c) Pre-image d) Codomain

II. Which of them is odd function

- a) x^2 b) x^3 c) $\sin x$ d) Both b and c

III. $\lim_{x \rightarrow 0} (1 + 4x)^{\frac{1}{x}} =$

- a) e^2 b) e^3 c) e d) e^4

IV. The order of complex number

- a) Exist b) does not exist c) define d) un define

V. If $f(x) = \sqrt{5x}$ is a function then $f'(x)$

- a) $\frac{1}{\sqrt{5x}}$ b) $\sqrt{\frac{5}{x}}$ c) $\frac{1}{\sqrt{x}}$ d) $\frac{5}{\sqrt{x}}$

VI. Derivative of constant function

- a) 0 b) 1 c) x d) ∞



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Instructor	Shumaila Asghar				
Exam Type	Mid Term				
Program	BSIT				
Session	Fall 2025				
Semester	2 nd				
Time Allowed	80 minutes				
Maximum Marks	24				
Student's Name		Roll No.		Section	
CLO	Question No	Marks		Obtained	
01	1,3,4	22			
02	2	8			

Instructor's Signatures: _____

Course Moderator/HOD's Signatures: _____

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(Subjective)

Note: Attempt each question according to the marks assigned to it.

Question 2:

The distance x metre gone by a body in t seconds is given by

$$x = t^3 - 3t^2 + 4$$

where t is time in seconds.

- i. Find the velocity at $t=4$
- ii. Find the acceleration at $t=4$

Question 3:

What is limit at infinity. Is it possible to having a finite number is a limit of a function whose independent variable approaches to Infinity solve the following expression and justify your answer?

$$\lim_{x \rightarrow \infty} \frac{2x^2 - 3x + 7}{3x^2 + 5x - 1}$$

Question 4:

Express $(1 - \sqrt{3}i)^3$ using De Moivre's Theorem